Malnormal contractions in every matrix dimension

Candidate full-proof packet

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Status. Candidate full proof; likely valid; human review recommended.

1 Source problem

For $A \in \mathbb{M}_n$, Mulcahy and Sinclair define

$$\operatorname{mal}(A) = \inf\left\{\|[A,B]\|_2 : B = B^*,\ \|B - \tau(B)I\|_2 = 1\right\},$$

where τ is normalized trace and $\|\cdot\|_2$ is the unnormalized Hilbert–Schmidt norm. A sequence is malnormal when its operator norms are uniformly bounded and its malnormality constants are uniformly positive.

The source constructs a universal $\kappa_0 > 0$ and contractions $X_m \in \mathbb{M}_{3m}$ for every sufficiently large m, with $\operatorname{mal}(X_m) \geq \kappa_0$ [1]. It then asks for a sequence in every dimension. The exact statement (PDF page 12) is reproduced below.

5. Further Problems

We list several other natural problems and conjectures related to malnormal matrices.

Problem 5.1. Construct a malnormal sequence of matrices A_n with $A_n \in \mathbb{M}_n$ for all $n = 1, 2, 3, \dots$.

We solve the problem by showing that malnormality can be padded upward one dimension at a time with quantitative control.

2 The padding lemma

The key point is to add a new scalar block outside the spectrum and couple it to the old space by one nonzero column. Different blocks of a small commutator then successively control the off-diagonal vector, the old self-adjoint block modulo scalars, and the difference between the two scalar parts.

Lemma 1 (one-dimensional padding)**.** *Let $A \in \mathbb{M}_m$ satisfy $\|A\| \leq M$ and $\operatorname{mal}(A) \geq \kappa > 0$. Choose any unit vector $v \in \mathbb{C}^m$, put $L = M + 1$, and define*

$$T = \begin{pmatrix} A & v \\ 0 & L \end{pmatrix} \in \mathbb{M}_{m+1}.$$

Then $\|T\| \leq M + 2$ and

$$\operatorname{mal}(T) \geq D(\kappa, M)^{-1}, \qquad D(\kappa, M)^2 = \left(\frac{2}{\kappa}\right)^2 + 2 + \left(2M + 2 + \frac{2}{\kappa}\right)^2.$$

Proof. The norm estimate follows by splitting T into its block diagonal and its upper-right rank-one block.

Let

$$B = \begin{pmatrix} C & x \\ x^* & b \end{pmatrix} = B^*, \qquad C \in \mathbb{M}_m, \quad x \in \mathbb{C}^m, \quad b \in \mathbb{R},$$

and set $\varepsilon = \|[T, B]\|_2$. Direct multiplication gives

$$[T, B] = \begin{pmatrix} [A, C] + vx^* & (A - LI)x + vb - Cv \\ x^*(LI - A) & -x^*v \end{pmatrix}.$$

Each block has Hilbert–Schmidt norm at most ε. Since $L - \|A\| = 1$, the lower-left block gives

$$\|x\|_2 \leq \|x^*(LI - A)\|_2 \leq \varepsilon.$$

The upper-left block now yields

$$\|[A, C]\|_2 \leq \varepsilon + \|vx^*\|_2 \leq 2\varepsilon.$$

Writing $c = \tau(C)$ and using $\mathrm{mal}(A) \geq \kappa$, we obtain

$$\|C - cI\|_2 \leq \frac{2}{\kappa}\varepsilon.$$

Finally, the upper-right block can be rewritten as

$$(A - LI)x + v(b - c) - (C - cI)v.$$

Because $\|v\| = 1$, it follows that

$$\begin{aligned} |b - c| &\leq \varepsilon + (\|A\| + L)\|x\|_2 + \|(C - cI)v\|_2 \\ &\leq \left(2M + 2 + \frac{2}{\kappa}\right)\varepsilon. \end{aligned}$$

Consequently,

$$\begin{aligned} \|B - cI_{m+1}\|_2^2 &= \|C - cI_m\|_2^2 + 2\|x\|_2^2 + |b - c|^2 \\ &\leq D(\kappa, M)^2\varepsilon^2. \end{aligned}$$

The normalized trace scalar is the Hilbert–Schmidt best scalar approximation, so

$$\|B - \tau(B)I_{m+1}\|_2 \leq \|B - cI_{m+1}\|_2.$$

This proves the malnormality estimate. □

Corollary 1 (padding contractions)**.** *If A is a contraction with* $\mathrm{mal}(A) \geq \kappa$*, then for any unit vector v the matrix*

$$P_v(A) = \frac{1}{3}\begin{pmatrix} A & v \\ 0 & 2 \end{pmatrix}$$

is a contraction in the next dimension and satisfies

$$\mathrm{mal}(P_v(A)) \geq F(\kappa), \qquad F(\kappa) = \frac{1}{3D(\kappa, 1)} > 0.$$

Proof. Apply Lemma 1 with $M = 1$ and divide the resulting matrix by three. Both the operator norm and the commutator scale linearly. □

3 A malnormal contraction in every dimension

Theorem 1. *There are a universal constant $\kappa_* > 0$ and contractions $A_n \in \mathbb{M}_n$ for every $n \geq 1$ such that*

$$\|[A_n, B]\|_2 \geq \kappa_* \|B - \tau(B) I_n\|_2 \qquad (B = B^* \in \mathbb{M}_n).$$

Thus Problem 5.1 of [1] has an affirmative answer.

Proof. Let $X_m \in \mathbb{M}_{3m}$ be the source's contractions for all $m \geq m_0$, with $\mathrm{mal}(X_m) \geq \kappa_0$. Put

$$\kappa_1 = F(\kappa_0), \qquad \kappa_2 = F(\kappa_1), \qquad \kappa_{\mathrm{large}} = \min\{\kappa_0, \kappa_1, \kappa_2\} > 0.$$

For $n = 3m + r \geq 3m_0$, where $r \in \{0, 1, 2\}$, start with X_m and apply Corollary 1 exactly r times, choosing an arbitrary unit padding vector at each step. The resulting matrix lies in $\mathbb{M}_n$, is a contraction, and has malnormality at least κ_{large}.

Only finitely many n remain. For $n \geq 2$, let S_n be the nilpotent unilateral shift, a single Jordan block with ones on the superdiagonal. Its commutant consists of upper triangular Toeplitz matrices, and a self-adjoint upper triangular matrix is diagonal. Hence every self-adjoint matrix commuting with S_n is scalar. Compactness of the unit sphere in the finite-dimensional space of trace-zero self-adjoint matrices gives $\mathrm{mal}(S_n) > 0$. For $n = 1$ the required inequality is vacuous. Taking the minimum of κ_{large} and these finitely many positive constants produces $\kappa_* > 0$ and completes the sequence. □

Corollary 2 (bounded-gap principle)**.** *Suppose uniformly malnormal contractions exist in an unbounded set of matrix dimensions whose successive gaps are bounded by q. Then uniformly malnormal contractions exist in every sufficiently large dimension.*

Proof. Apply Corollary 1 at most $q - 1$ times. The finite list $\kappa, F(\kappa), \dots, F^{q-1}(\kappa)$ has a positive minimum. □

4 Scope and literature boundary

The construction inherits the nonconstructive choice of the source's quantum expanders; it resolves the dimension gap, not the separate request for a fully explicit algebraic seed. It also does not address the sharp constant, the random-matrix conjecture, or the proposed converse with quantum edge expanders.

A bounded novelty search on 11 August 2026 used the exact title and problem sentence, all-dimension malnormal sequences, one-dimensional block padding, and subsequent references to the paper. It found the source and unrelated group-theoretic uses of "malnormal," but no later solution of Problem 5.1 or matching padding lemma. Novelty confidence is moderate.

Human-review recommendation. Audit the four displayed blocks of $[T, B]$, especially the lower-left inverse bound and the shared scalar $c = \tau(C)$. Then check that the source's Proposition 3.3 indeed supplies its $3m$ family for every sufficiently large m, as used here.

References

[1] G. Mulcahy and T. Sinclair, *Malnormal matrices*, Proc. Amer. Math. Soc. **150** (2022), 2989–3001; arXiv:2009.11139.